_eda_price_2026_Q1

April 7, 2026

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import subprocess
import os
from scipy.stats import chi2_contingency
from datetime import timedelta

from etl.customer import aggregate_customer
from etl.subsidy import report_customer_conversion_by_subsidy_issue_status,␣
↪report_customer_metrics_by_subsidy_issue_status
from etl.subsidy import visualize_conversion_by_subsidy_issues,␣
↪report_customer_subsidy_issues_by_product
from etl.subsidy import visualize_customer_subsidy_issues_by_product,␣
↪visualize_policy_suspension_impact
from etl.price import visualize_conversion_by_price_sweet_spots,␣
↪visualize_conversion_by_price_sweet_spots_variant
from etl.price import report_product_conversion_by_price_quartile,␣
↪visualize_conversion_by_equipment_category_price
from etl.price import visualise_conversion_by_brand_price,␣
↪show_two_panel_conversion_by_brand_price
from etl.price import visualize_conversion_by_equipment_type_brand_price,␣
↪show_conversion_by_price_dashboard,␣
↪show_product_type_price_conversion_heatmap
from etl.season import visualize_conversion_by_season,␣
↪visualize_regional_seasonality
from etl.period import visualize_conversion_by_year,␣
↪report_best_and_worst_months, show_conversion_by_price_over_time,␣
↪overlay_all_years_on_one_plot
from etl.period import visualize_conversion_by_prices_sweet_spot_shift,␣
↪visualize_decision_times, show_decision_time_dashboard
from etl.product import visualise_heat_pump_performance,␣
↪show_product_type_by_brand_heatmap, show_product_type_by_agency_heatmap,␣
↪show_brand_by_product_type_heatmap
```

```python
import warnings
warnings.filterwarnings('ignore')

df_clean = pd.read_csv('cleaned_quote_data.csv')
df_clean['dt_creation_devis'] = pd.to_datetime(df_clean['dt_creation_devis'])
print(f"Loaded data: {len(df_clean):,} quotes from {df_clean['numero_compte'].
 ↪nunique():,} customers")
```

```
Loaded data: 37,008 quotes from 25,716 customers
```

```python
customers = aggregate_customer(df_clean)
```

```
Quote-level columns created:
  - out_of_pocket: 34,991 values
  - subsidy_issue: 9,076 issues (24.5%)
  - brand-product pairs: 146 unique combinations

================================================================================
BRAND-PRODUCT VALIDATION
================================================================================

  ATLANTIC + Stove customers (should be 0): 0
  FIXED: No ATLANTIC + Stove combinations found!

  Customer dataset created: 25,716 customers
  Total columns: 61
  Key columns now available:
    - main_brand: from brand-product pair (most common option)
    - main_equipment_category: from brand-product pair (most common option)
    - brand_product_nunique: 1.2 avg options per customer
    - shopped_around: 4,484 customers (17.4%)
    - decision_days: from customer_duration_days
    - price_cv: from std_out_of_pocket / avg_out_of_pocket
    - quote_count: alias for total_quotes
    - price_range: 1142 avg
    - price_volatility: 0.26 avg

Subsidy issue type distribution:
subsidy_issue_type
No Issues            18660
Both                  2480
CEE Only              2361
MaPrimeRénov Only     2215
Name: count, dtype: int64

  Saved to 'customer_master_data.csv'
```

```python
all_customers = pd.read_csv('customer_master_data.csv')
```

```python
# Using quarter column (if you have it)
customers_2026_q1 = all_customers[
    (all_customers['year'] == 2026) &
    (all_customers['quarter'] == 1)
]

# Using date range
customers_2026_q1 = all_customers[
    (all_customers['first_quote_date'] >= '2026-01-01') &
    (all_customers['first_quote_date'] <= '2026-03-31')
]

print(f"Q1 2026 customers: {len(customers_2026_q1):,}")
```

```
Q1 2026 customers: 2,397
```

```python
customers = customers_2026_q1
```

```python
conv_no_issue, conv_with_issue =␣
 ↪report_customer_conversion_by_subsidy_issue_status(customers)
report_customer_metrics_by_subsidy_issue_status(customers)
visualize_conversion_by_subsidy_issues(customers, conv_no_issue,␣
 ↪conv_with_issue)
visualize_policy_suspension_impact(customers, price_var='max_out_of_pocket')

top_products, no_issue_pct, issue_pct =␣
 ↪report_customer_subsidy_issues_by_product(customers)
visualize_customer_subsidy_issues_by_product(customers, top_products,␣
 ↪no_issue_pct, issue_pct)

visualize_conversion_by_price_sweet_spots(customers,␣
 ↪price_var='max_out_of_pocket')
visualize_conversion_by_price_sweet_spots_variant(customers,␣
 ↪price_var='max_out_of_pocket')
report_product_conversion_by_price_quartile(customers)
visualize_conversion_by_equipment_category_price(customers,␣
 ↪price_var='max_out_of_pocket')
visualise_conversion_by_brand_price(customers, price_var='max_out_of_pocket')
show_two_panel_conversion_by_brand_price(customers,␣
 ↪price_var='max_out_of_pocket')
visualize_conversion_by_equipment_type_brand_price(customers,␣
 ↪price_var='max_out_of_pocket')

visualize_conversion_by_season(customers, price_var='max_out_of_pocket')
visualize_regional_seasonality(customers, price_var='max_out_of_pocket')
```

```
visualize_conversion_by_year(customers, price_var='max_out_of_pocket')
report_best_and_worst_months(customers, price_var='max_out_of_pocket')
show_conversion_by_price_over_time(customers, price_var='max_out_of_pocket')
overlay_all_years_on_one_plot(customers, price_var='max_out_of_pocket')
visualize_conversion_by_prices_sweet_spot_shift(customers,␣
 ↪price_var='max_out_of_pocket')
visualize_decision_times(customers, price_var='max_out_of_pocket')
show_product_type_price_conversion_heatmap(customers,␣
 ↪price_var='max_out_of_pocket')
show_decision_time_dashboard(customers, price_var='max_out_of_pocket')

visualise_heat_pump_performance(customers, price_var='max_out_of_pocket')

show_conversion_by_price_dashboard(customers, price_var='max_out_of_pocket')

show_product_type_by_brand_heatmap(customers)
show_product_type_by_agency_heatmap(customers)
show_brand_by_product_type_heatmap(customers)
```

```
Customer conversion by subsidy issue status:

                   conversion_rate  customer_count  converted_count
had_subsidy_issue
False                     0.256855            1787            459.0
True                      0.285246             610            174.0

Chi-square p-value: 1.867727e-01
Statistically significant: NO

Conversion without issues: 25.69%
Conversion with issues: 28.52%
Difference: 2.8 percentage points
Customer-level finding: +2.8 points

================================================================================
Customer characteristics by subsidy issue status
================================================================================

Metric comparison:

total_quotes:
  No issues: 1.36
  With issues: 1.47
  Ratio: 1.09x

avg_quote_amount:
```

```
  No issues: 5717.54
  With issues: 9724.94
  Ratio: 1.70x

max_quote_amount:
  No issues: 6037.00
  With issues: 10637.81
  Ratio: 1.76x

customer_duration_days:
  No issues: 2.15
  With issues: 3.01
  Ratio: 1.40x

price_range:
  No issues: 698.12
  With issues: 1805.23
  Ratio: 2.59x

multiple_quotes:
  No issues: 0.27
  With issues: 0.34
  Ratio: 1.24x

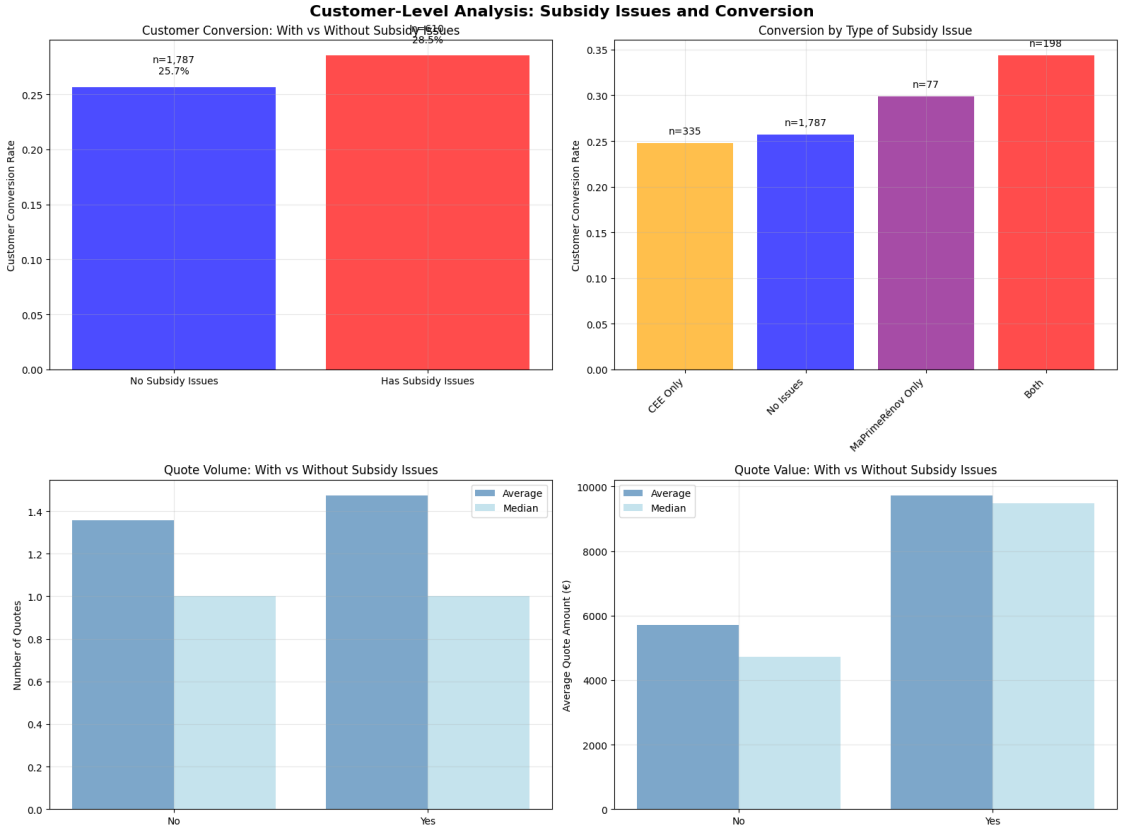
Quote pattern distribution by subsidy issue:
had_subsidy_issue  False  True
quote_pattern
Single quote       0.726  0.659
Three+ quotes      0.059  0.093
Two quotes         0.215  0.248

Decision speed by subsidy issue:
had_subsidy_issue   False  True
decision_speed
Long (>30 days)     0.011  0.025
Medium (8-30 days)  0.035  0.044
Quick (2-7 days)    0.055  0.090
Same day decision   0.899  0.841

Product family by subsidy issue status:
had_subsidy_issue                 False  True
main_product_family
Accessoire de pose                0.002  0.000
Appareil hybride                  0.001  0.003
Autres                            0.008  0.002
Chaudière                         0.427  0.056
Climatisation                     0.217  0.139
ECS : Chauffe-eau ou adoucisseur  0.074  0.010
```

```
Emetteur de chauffage  ou chappe  0.005  0.008
Photovoltaïque                    0.016  0.011
Plomberie Sanitaire               0.001  0.000
Pompe à chaleur                   0.055  0.451
Poêle                             0.181  0.320
Produit VMC                       0.004  0.000
Unknown                           0.009  0.000

Conversion by subsidy issue type:
                        mean  count
subsidy_issue_type
Both                0.343434    198
MaPrimeRénov Only   0.298701     77
No Issues           0.256855   1787
CEE Only            0.247761    335
```

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================================================================================
Policy Suspension Impact
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Analyzing 2,310 customers
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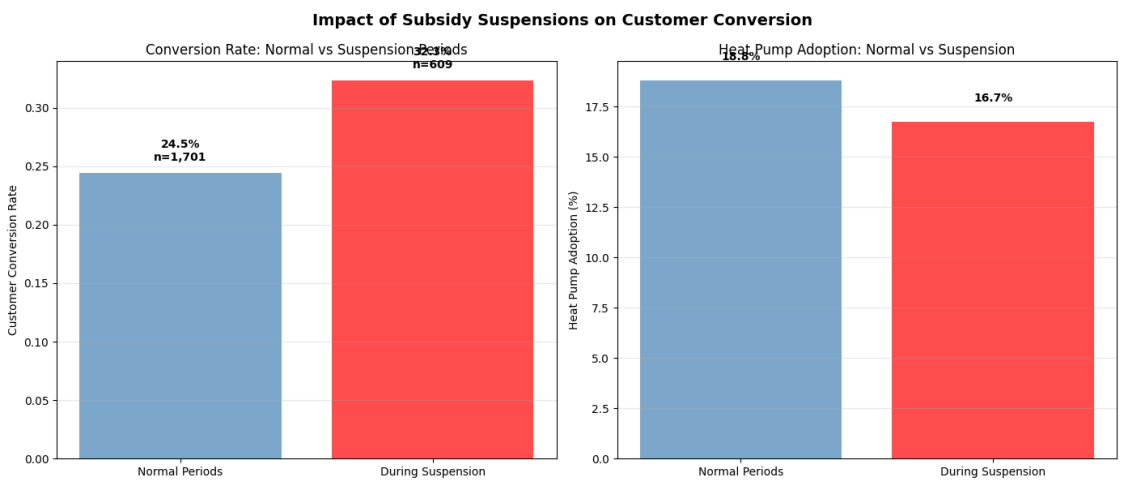
Price range: €1,327 to €25,676
Median price: €6,126
Unique values in during_suspension: [False  True]

Conversion during subsidy suspensions:
                       mean  count
Normal Periods     0.244562   1701
During Suspension  0.323481    609

Suspension impact p-value: 0.0002
Statistically significant: YES

Product mix during suspensions:
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove
during_suspension
False                    22.1    31.1       16.6    8.9   21.3
True                     13.5    38.6       14.8    8.4   24.8
```



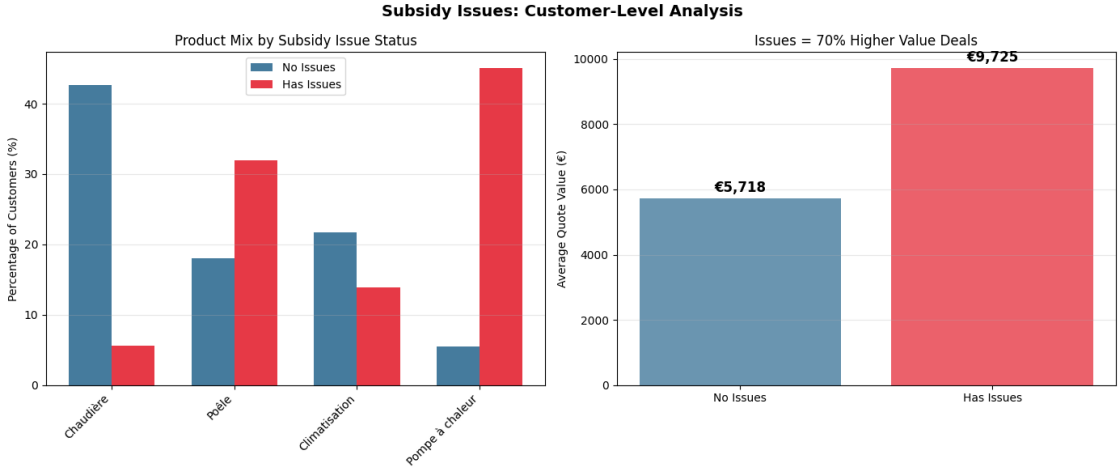

```
Unique values in main_product_family:
main_product_family
Chaudière                          797
Poêle                              518
Climatisation                      472
Pompe à chaleur                    373
ECS : Chauffe-eau ou adoucisseur   139
Photovoltaïque                      36
Unknown                             16
Autres                              16
Emetteur de chauffage  ou chappe    14
Produit VMC                          8
Appareil hybride                     3
```

```
Accessoire de pose                    3
Plomberie Sanitaire                   2
Name: count, dtype: int64

Top products:
main_product_family
Chaudière                           797
Poêle                               518
Climatisation                       472
Pompe à chaleur                     373
ECS : Chauffe-eau ou adoucisseur    139
Photovoltaïque                       36
Unknown                              16
Autres                               16
Emetteur de chauffage  ou chappe     14
Produit VMC                           8
Name: count, dtype: int64

Using these products: ['Chaudière', 'Poêle', 'Climatisation', 'Pompe à chaleur']

No Issue %: [np.float64(42.697257974258534), np.float64(18.074986010072745),
np.float64(21.65640738668159), np.float64(5.484051482932289)]
Has Issue %: [np.float64(5.573770491803279), np.float64(31.967213114754102),
np.float64(13.934426229508196), np.float64(45.08196721311475)]
```

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==============================================================================
Threshold effect
==============================================================================

Analyzing 2,310 customers
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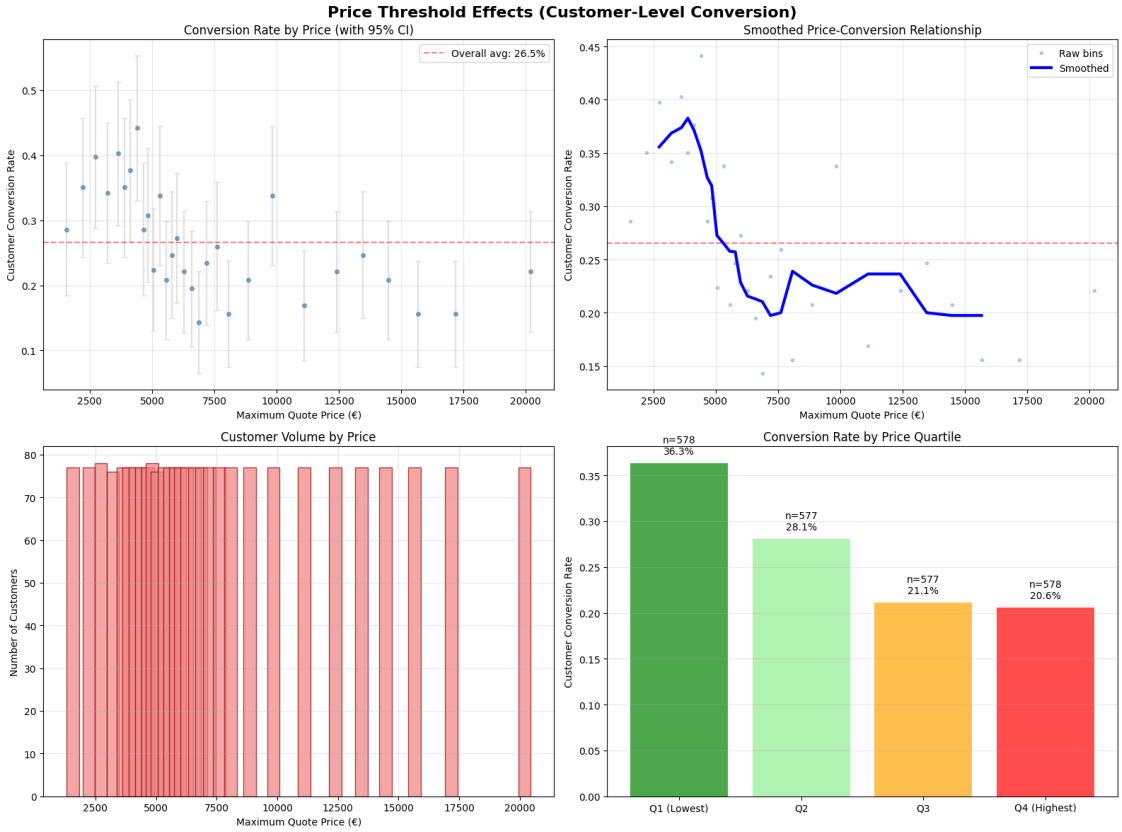
Price range: €1,327 to €25,676
Median price: €6,126
```

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================================================================================
THRESHOLD EFFECT: KEY FINDINGS
================================================================================

Conversion by Price Quartile:
  Q1 (Lowest): 36.3% (n=578)
  Q2: 28.1% (n=577)
  Q3: 21.1% (n=577)
  Q4 (Highest): 20.6% (n=578)

Difference between lowest and highest quartile: 15.7 percentage points

Sweet spots (above-average conversion):
  €1,567: 28.6% (n=77)
  €2,224: 35.1% (n=77)
  €2,728: 39.7% (n=78)
  €3,217: 34.2% (n=76)
  €3,624: 40.3% (n=77)
```

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  €3,880: 35.1% (n=77)
  €4,128: 37.7% (n=77)
  €4,404: 44.2% (n=77)
  €4,658: 28.6% (n=77)
  €4,838: 30.8% (n=78)
  €5,327: 33.8% (n=77)
  €5,993: 27.3% (n=77)
  €9,838: 33.8% (n=77)

Analyzing 2,310 customers
Price range: €1,327 to €25,676
Median price: €6,126

================================================================================
Product mix by price quartile
================================================================================

Product distribution by price quartile (%):
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove
price_quartile
Q1_Low                   20.2    46.9        0.0   27.1    5.8
Q2_MidLow                15.4    44.0        0.2    2.9   37.5
Q3_MidHigh               23.4    30.4        2.9    3.2   40.1
Q4_High                  20.2     9.5       61.5    4.7    4.1

================================================================================
Conversion rates by product and price quartile
================================================================================

Heat Pump (n=402):
  Q1_Low: nan%
  Q2_MidLow: 100.0%
  Q3_MidHigh: 35.3%
  Q4_High: 22.3%

Boiler (n=771):
  Q1_Low: 40.1%
  Q2_MidLow: 30.5%
  Q3_MidHigh: 26.8%
  Q4_High: 17.9%

AC (n=470):
  Q1_Low: 27.7%
  Q2_MidLow: 30.8%
  Q3_MidHigh: 20.3%
  Q4_High: 16.0%

Stove (n=530):
```

```
  Q1_Low: 52.9%
  Q2_MidLow: 25.3%
  Q3_MidHigh: 15.7%
  Q4_High: 16.7%

================================================================================
HEAT PUMP CUSTOMERS (n=454)
================================================================================
Average price: €15,574
Conversion rate: 22.7%
Subsidy issue rate: 75.1%

Heat pump adoption by quartile:
  Q1_Low: 0.3%
  Q2_MidLow: 0.2%
  Q3_MidHigh: 2.9%
  Q4_High: 70.0%

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Equipment Category Segment Effect
================================================================================

Analyzing 2,310 customers
Price range: €1,327 to €25,676
Median price: €6,126
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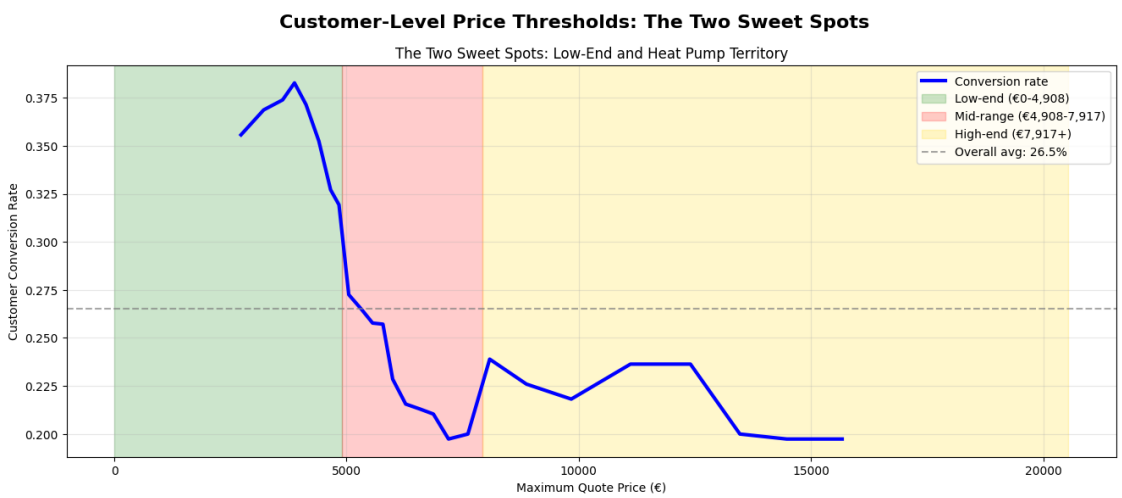

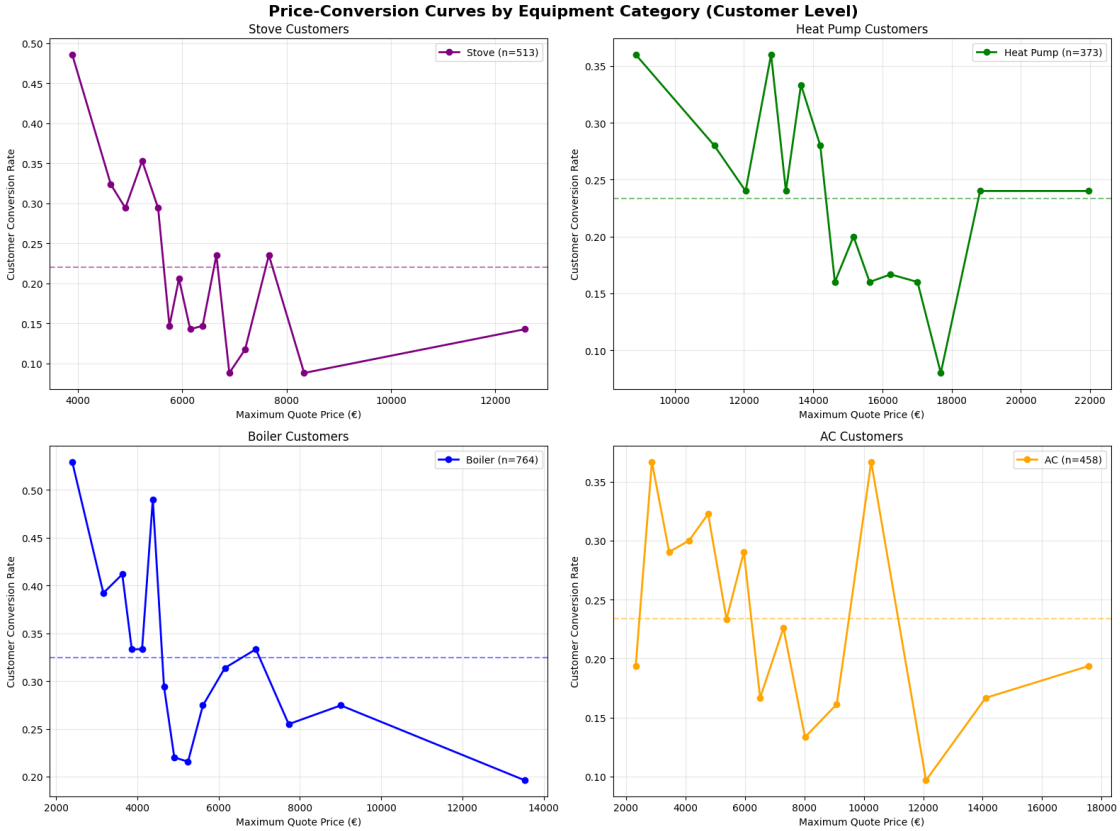


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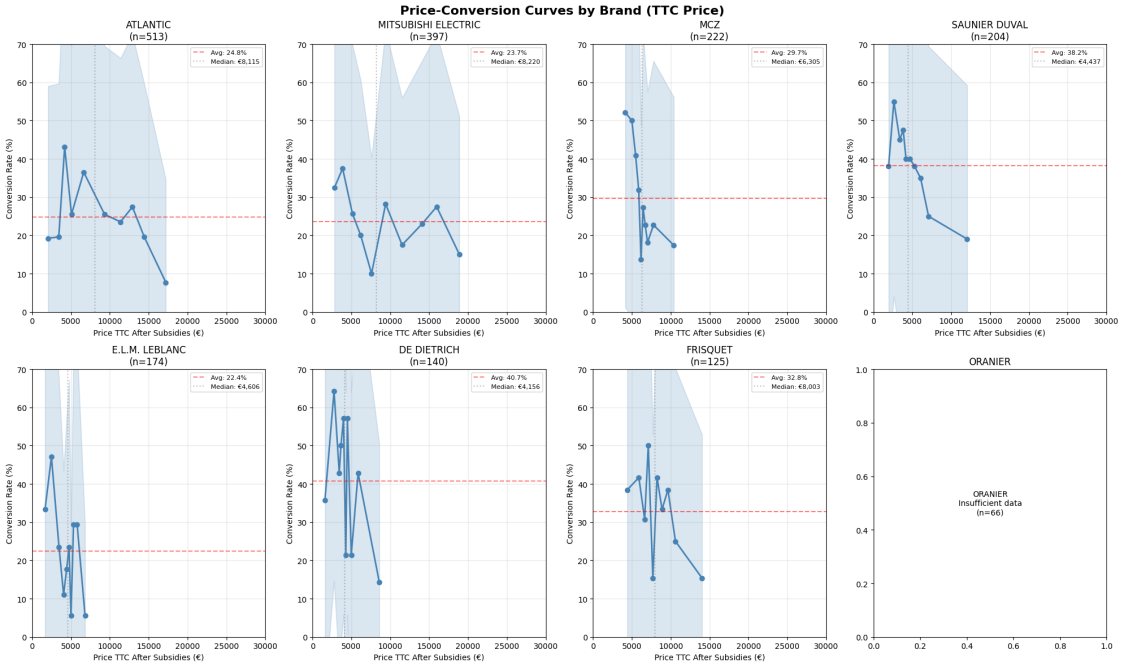
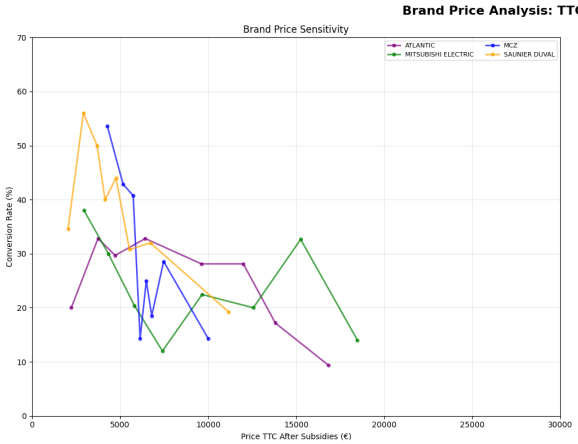
================================================================================
Conversion by Brand Price
================================================================================

Analyzing 2,310 customers
Price range: €1,327 to €25,676
Median price: €6,126
Top 8 brands: ['ATLANTIC', 'MITSUBISHI ELECTRIC', 'MCZ', 'SAUNIER DUVAL',
'E.L.M. LEBLANC', 'DE DIETRICH', 'FRISQUET', 'ORANIER']
```




```
Analyzing 2,310 customers
Price range: €1,327 to €25,676
Median price: €6,126
```



Brand Performance Summary

Brand	Customers	Avg Conv	Avg Price	Best Price Range	Best Conv
ATLANTIC	513	24.8%	€8,658	€3,812-5,756	34.3%
MITSUBISHI ELECTRIC	397	23.7%	€9,550	€2,043-4,521	35.0%
MCZ	222	29.7%	€6,512	€1,800-5,258	51.1%
SAUNIER DUVAL	204	38.2%	€5,130	€1,327-2,969	46.3%
E.L.M. LEBLANC	174	22.4%	€4,374	€1,342-2,787	40.0%
DE DIETRICH	140	40.7%	€4,359	€1,364-3,285	50.0%
FRISQUET	125	32.8%	€8,323	€2,391-6,452	40.0%

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================================================================================
TTC Price-conversion curves: 4 equipment types, multiple brands per graph
================================================================================

Analyzing 2,310 customers
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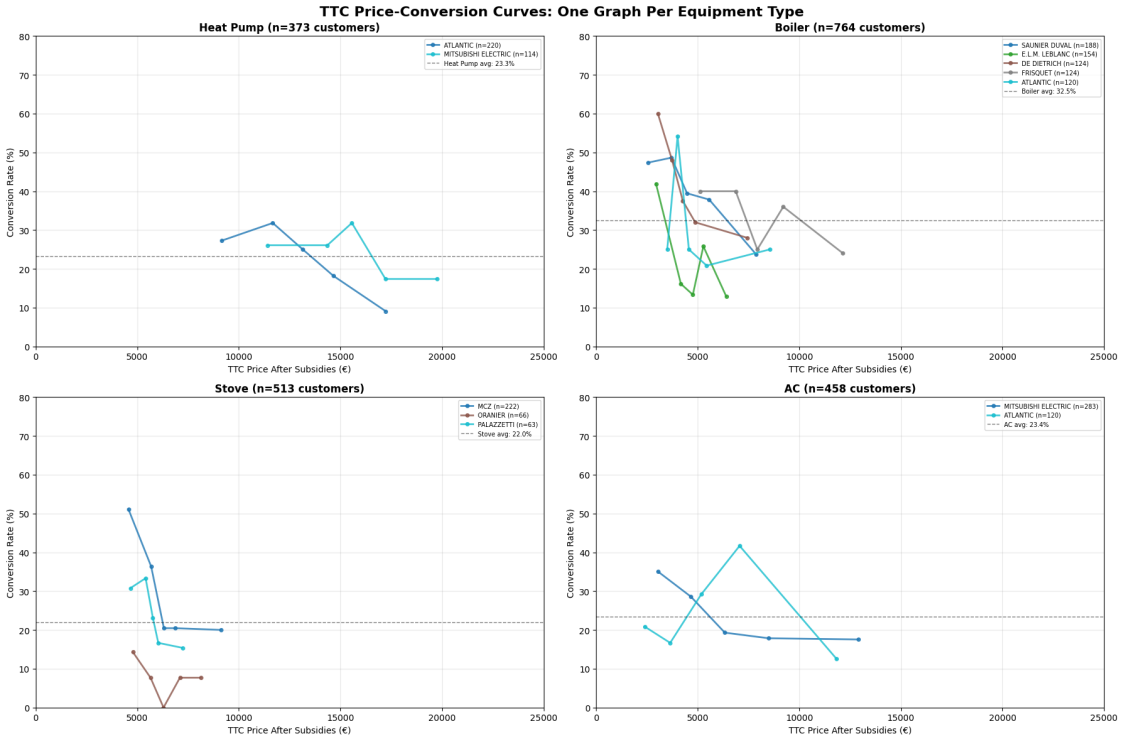
```
Price range: €1,327 to €25,676
Median price: €6,126

Heat Pump - Top brands: ['ATLANTIC', 'MITSUBISHI ELECTRIC']

Boiler - Top brands: ['SAUNIER DUVAL', 'E.L.M. LEBLANC', 'DE DIETRICH',
'FRISQUET', 'ATLANTIC']

Stove - Top brands: ['MCZ', 'ORANIER', 'PALAZZETTI']

AC - Top brands: ['MITSUBISHI ELECTRIC', 'ATLANTIC']
```

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================================================================================
Seasonal Effects on Customer Conversion
================================================================================
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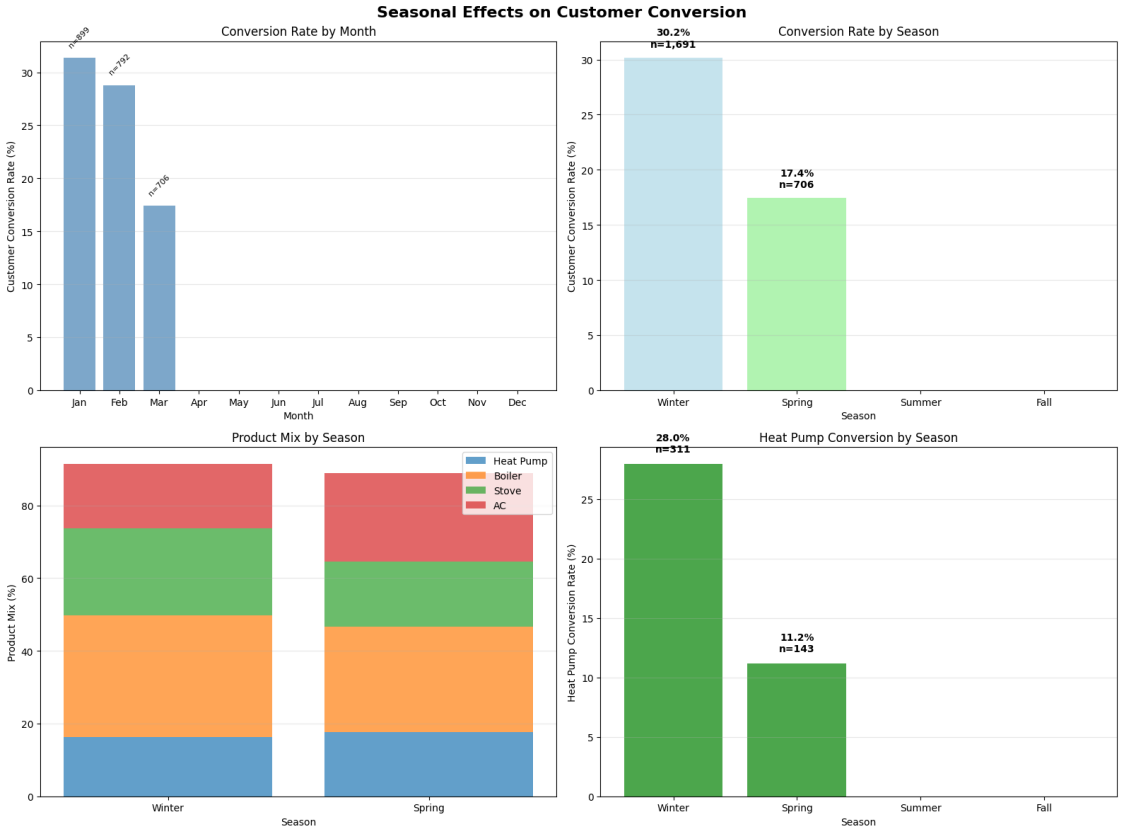


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================================================================================
SEASONAL STATISTICS
================================================================================

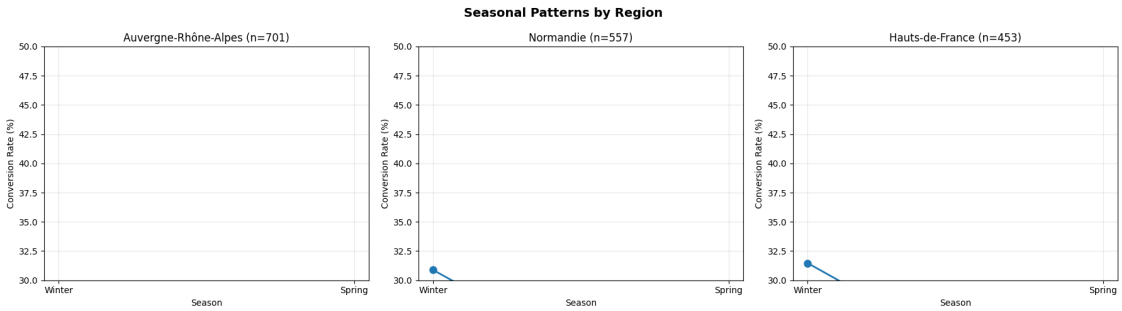
Conversion by season:
  Winter: 30.2% (n=1,691)
  Spring: 17.4% (n=706)

Seasonality chi-square p-value: 0.0000
Statistically significant: YES

================================================================================
Regional Seasonal Pattern
================================================================================

Analyzing 2,310 customers
Price range: €1,327 to €25,676
Median price: €6,126

Top regions: ['Auvergne-Rhône-Alpes', 'Normandie', 'Hauts-de-France']
```

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================================================================================
Year-over-year trends
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Analyzing 2,310 customers
Price range: €1,327 to €25,676
Median price: €6,126

Quarterly conversion rates:
                  mean  count
year_quarter
2026Q1        0.265368   2310
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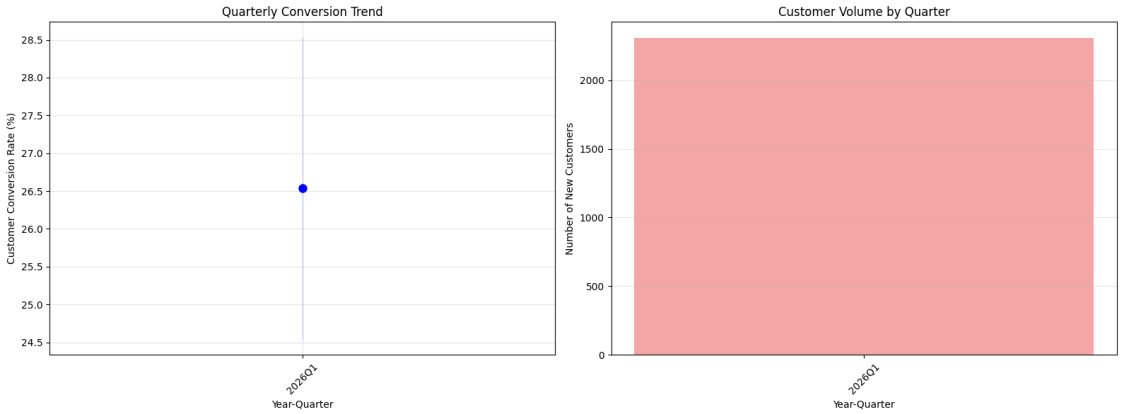


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================================================================================
Best and Worst Months Report
================================================================================

Analyzing 2,310 customers
Price range: €1,327 to €25,676
```

```
Median price: €6,126

Conversion during subsidy suspensions:
                       mean  count
Normal Periods     0.244562   1701
During Suspension  0.323481    609

Suspension impact p-value: 0.0002
Statistically significant: YES

Best month: 1 (31.7%, n=871)

Best season: Winter (30.2%)
Worst season: Spring (17.8%)

Heat pump conversion by season:
  Winter: 29.1%
  Spring: 12.0%

Analyzing 2,310 customers
Price range: €1,327 to €25,676
Median price: €6,126
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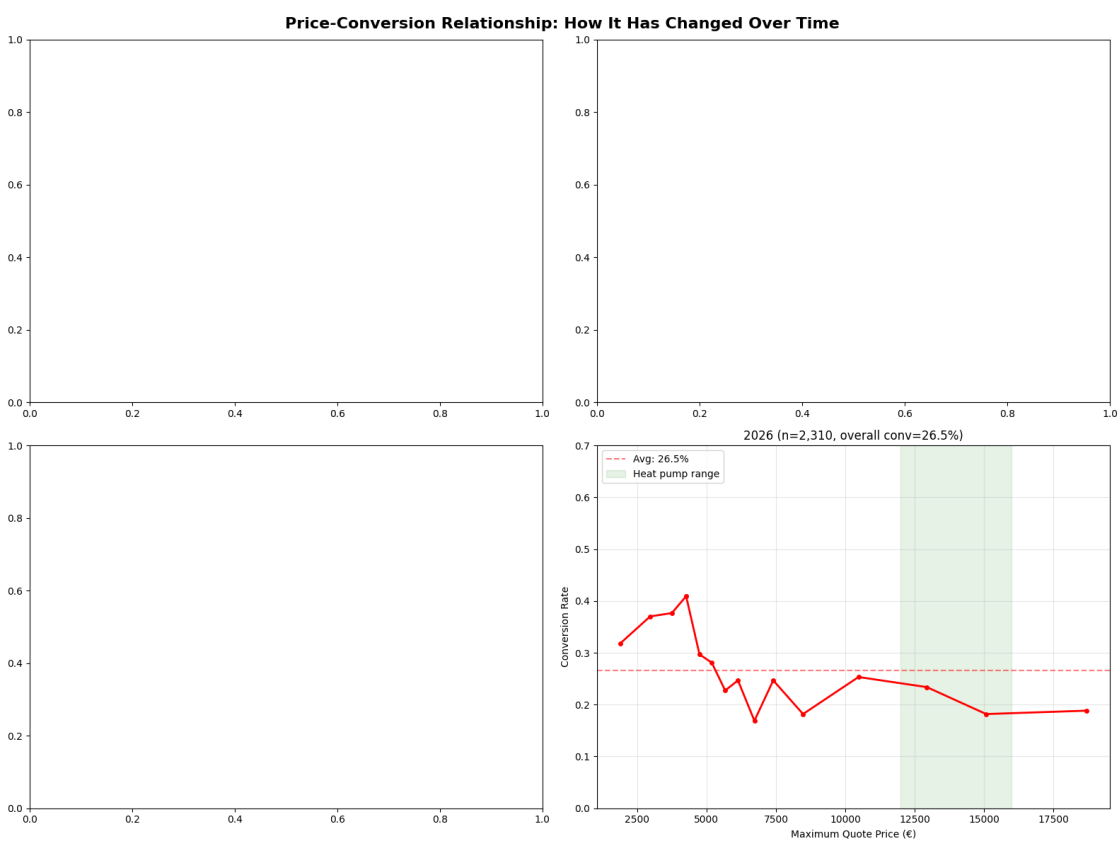

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Analyzing 2,310 customers
Price range: €1,327 to €25,676
Median price: €6,126
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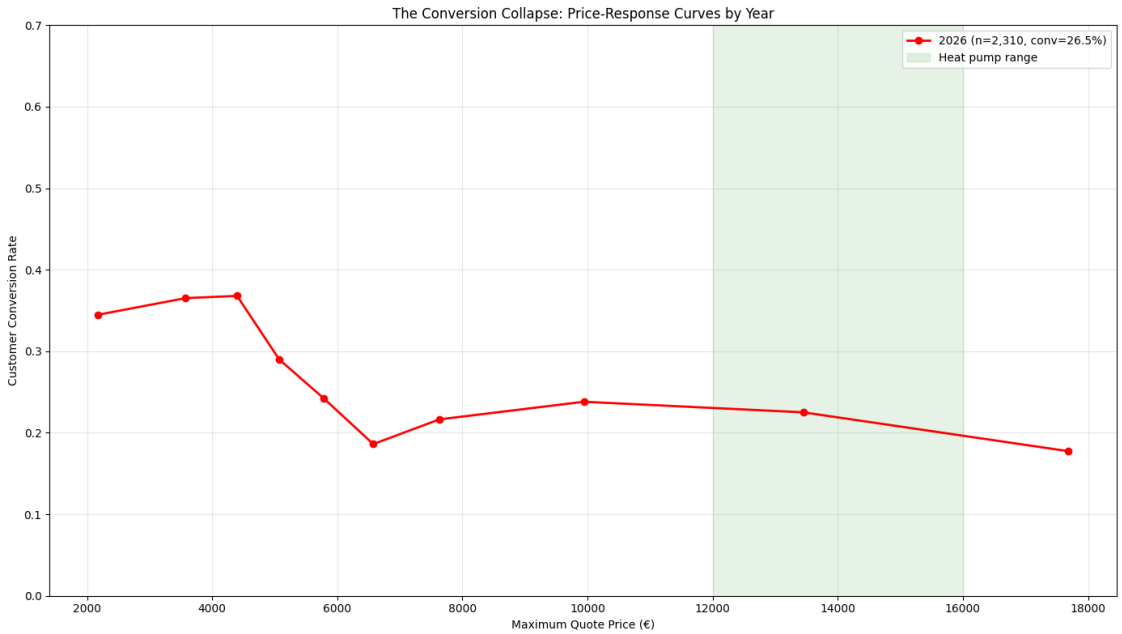


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================================================================================
Sweet Spot Analysis Over Time
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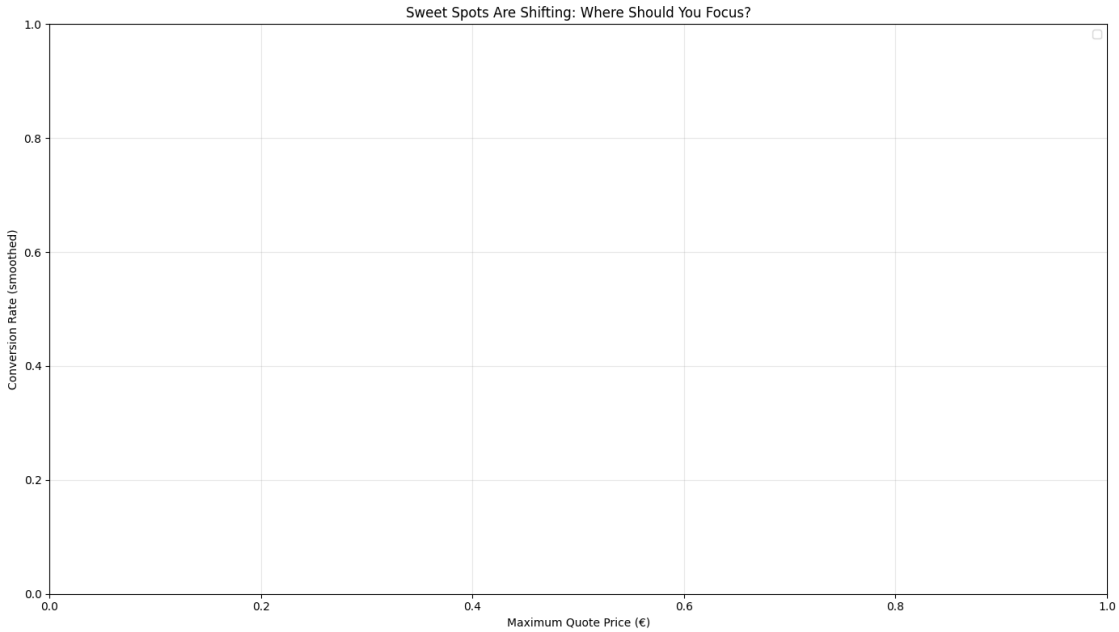
Analyzing 2,310 customers
Price range: €1,327 to €25,676
Median price: €6,126

Sweet spots by year (price points with conversion > avg + 3%):
```

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================================================================================
Administrative Uncertainty Effect
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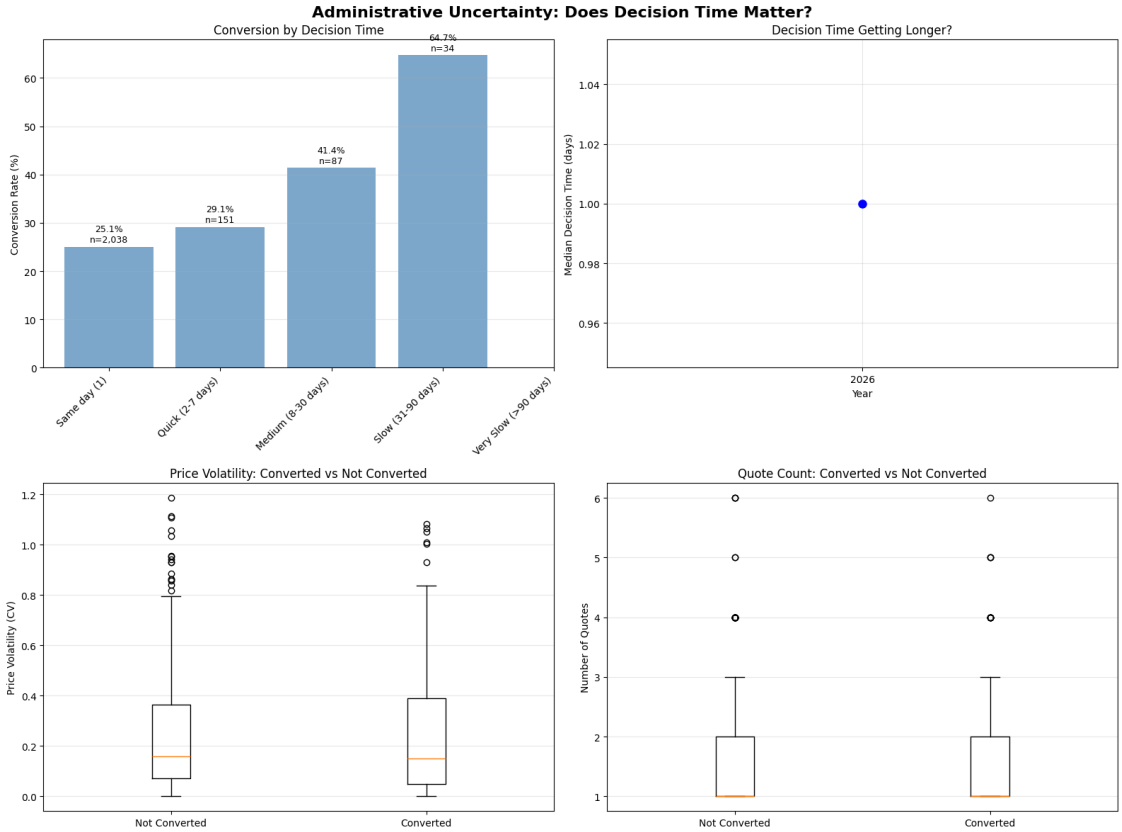
Analyzing 2,310 customers
Price range: €1,327 to €25,676
Median price: €6,126

Decision time distribution:
count    2310.000000
mean        2.386580
std         6.382556
min         1.000000
25%         1.000000
50%         1.000000
75%         1.000000
max        67.000000
Name: decision_days, dtype: float64

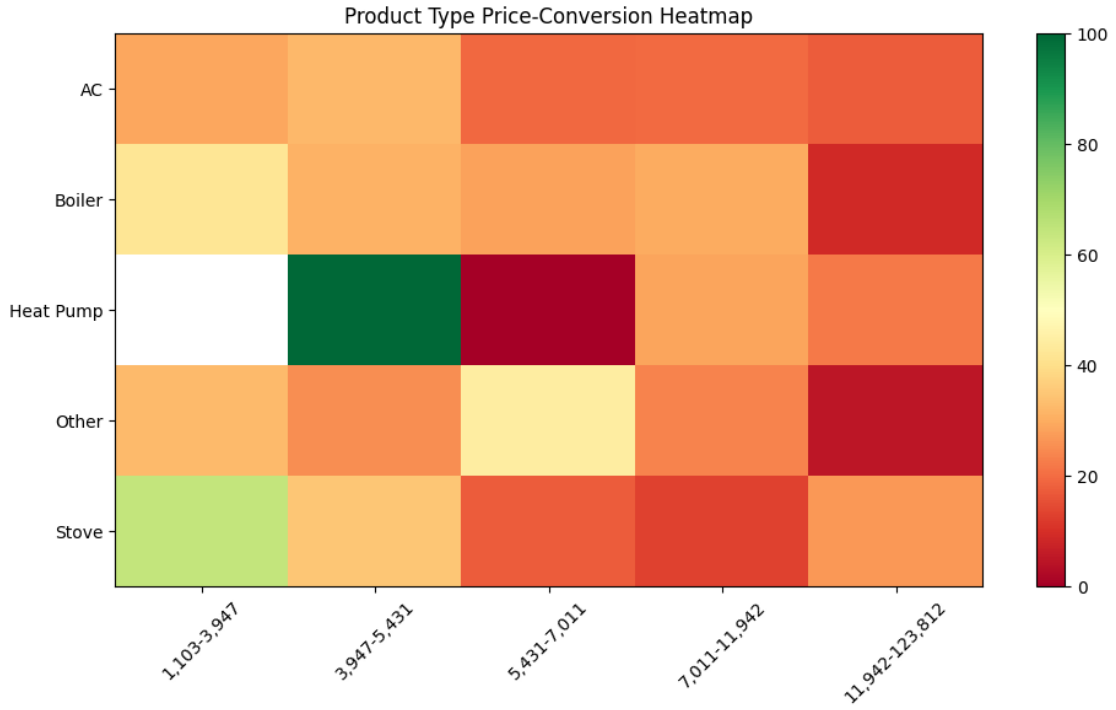
Customers with 1-day decision: 2,038 (88.2%)
```




```
================================================================================
Product Type - Price Conversion Heatmap
================================================================================
Simple pivot table:
price_bin_simple         (1103.269, 3947.814]  (3947.814, 5431.284]  \
main_equipment_category
AC                                  29.126214             32.352941
Boiler                              41.919192             31.034483
Heat Pump                                 NaN            100.000000
Other                               32.484076             25.000000
Stove                               64.285714             34.883721

price_bin_simple         (5431.284, 7011.9]  (7011.9, 11942.726]  \
main_equipment_category
AC                                19.512195            19.548872
Boiler                            28.358209            29.861111
Heat Pump                          0.000000            28.571429
Other                             44.444444            23.529412
Stove                             17.372881            13.223140
```

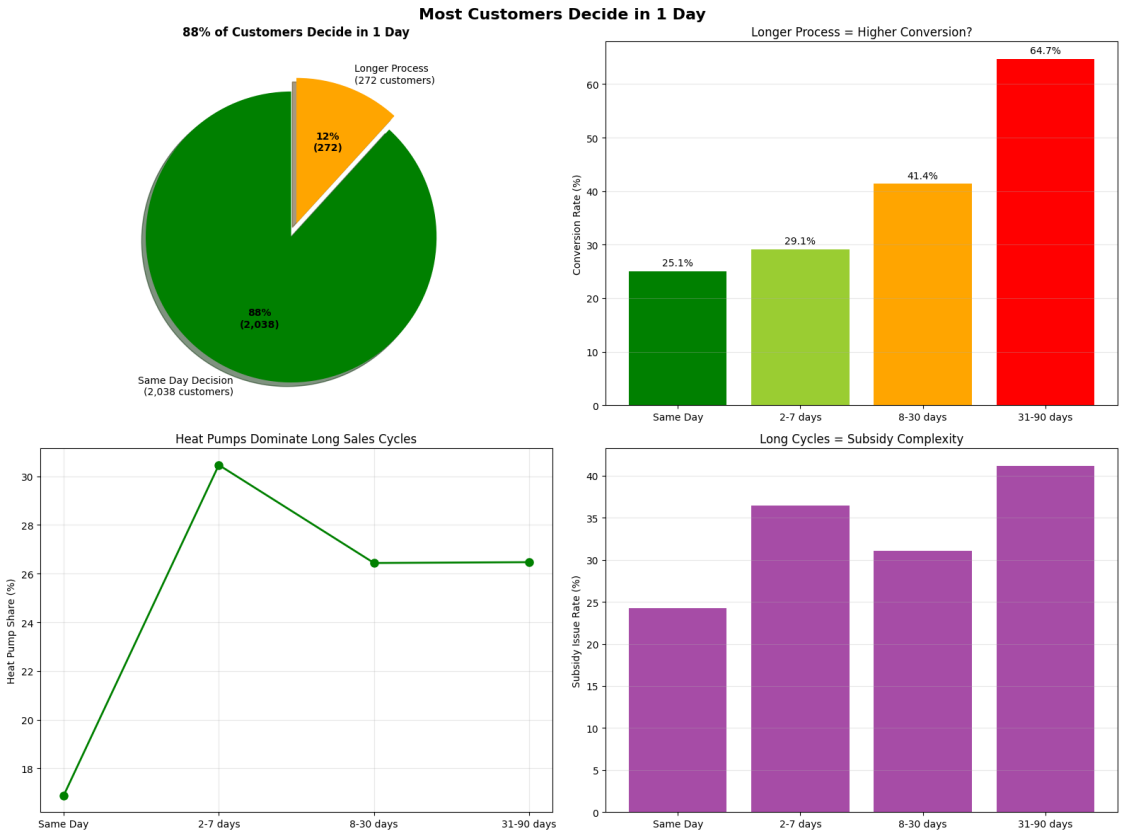
```
price_bin_simple         (11942.726, 123812.12]
main_equipment_category
AC                                     17.283951
Boiler                                  8.823529
Heat Pump                              22.049689
Other                                   5.000000
Stove                                  26.666667
```




```
================================================================================
CHECKING MAIN PRODUCT FAMILY
================================================================================
main_product_family
Chaudière                           797
Poêle                               518
Climatisation                       472
Pompe à chaleur                     373
ECS : Chauffe-eau ou adoucisseur    139
Photovoltaïque                       36
Unknown                              16
Autres                               16
Emetteur de chauffage  ou chappe     14
Produit VMC                           8
Appareil hybride                      3
Accessoire de pose                    3
```

```
Plomberie Sanitaire                    2
Name: count, dtype: int64

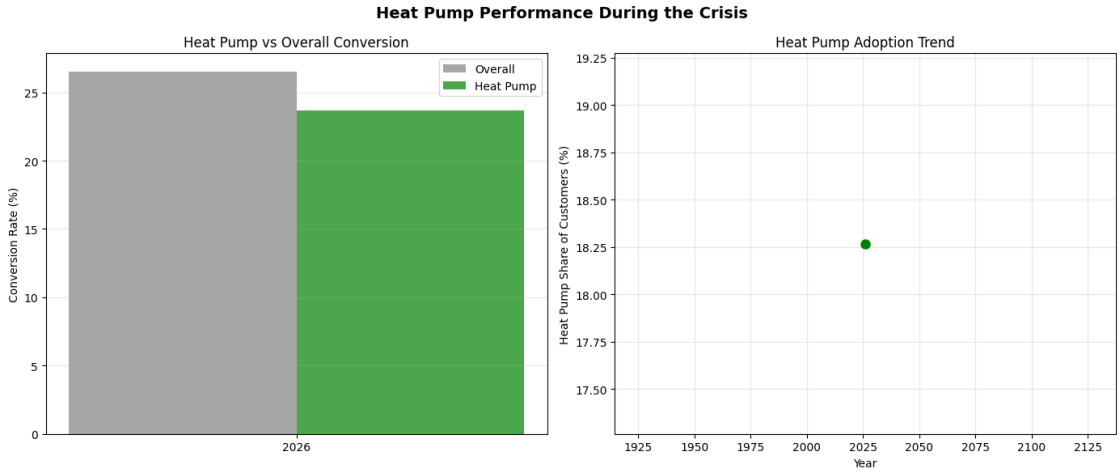
Analyzing 2,310 customers
Price range: €1,327 to €25,676
Median price: €6,126
```




```
================================================================================
Heat Pump Performance Over Time
================================================================================

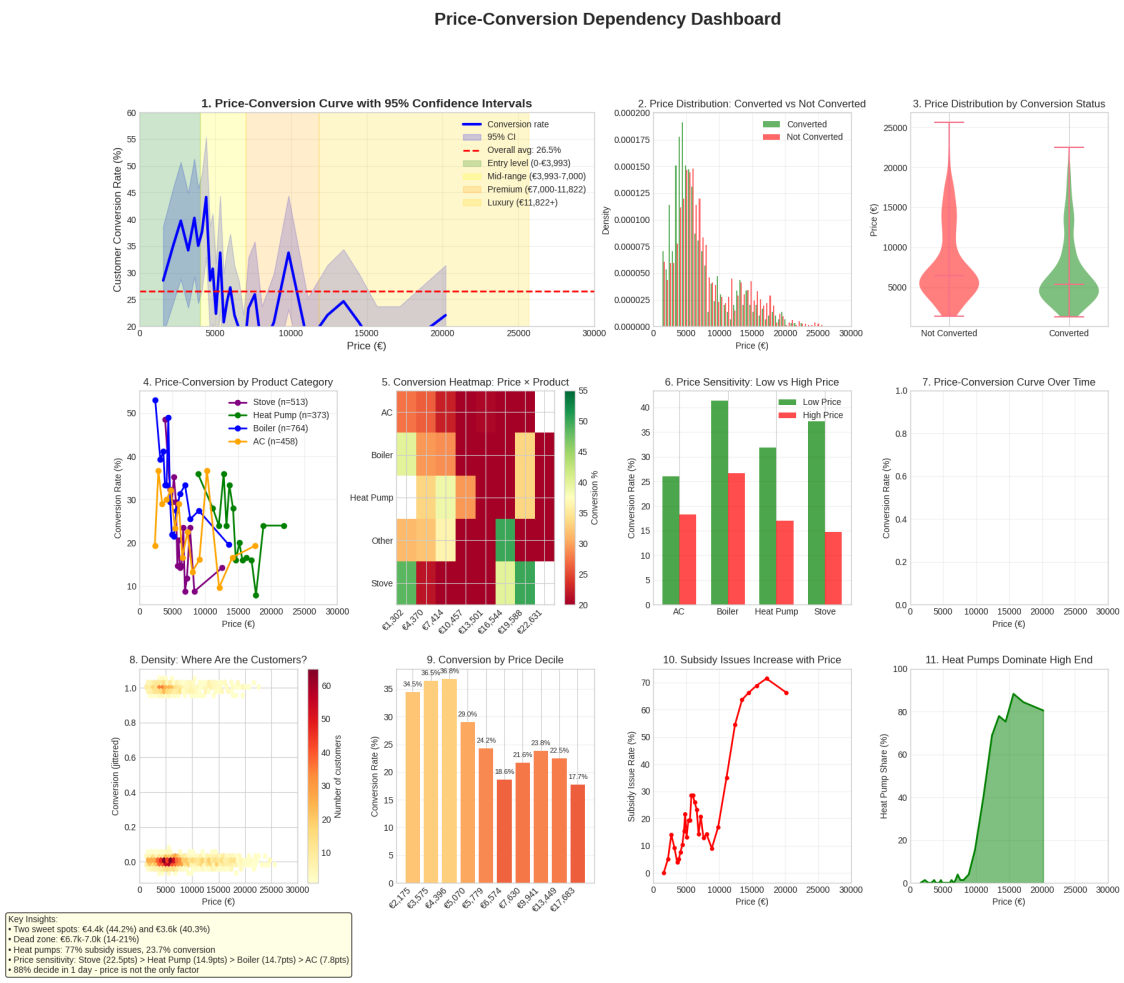
Analyzing 2,310 customers
Price range: €1,327 to €25,676
Median price: €6,126

Heat pump customers by year:
        conversion_rate  customer_count  avg_price  subsidy_issue_rate
period
2026              0.237             422  14897.542               0.775
```

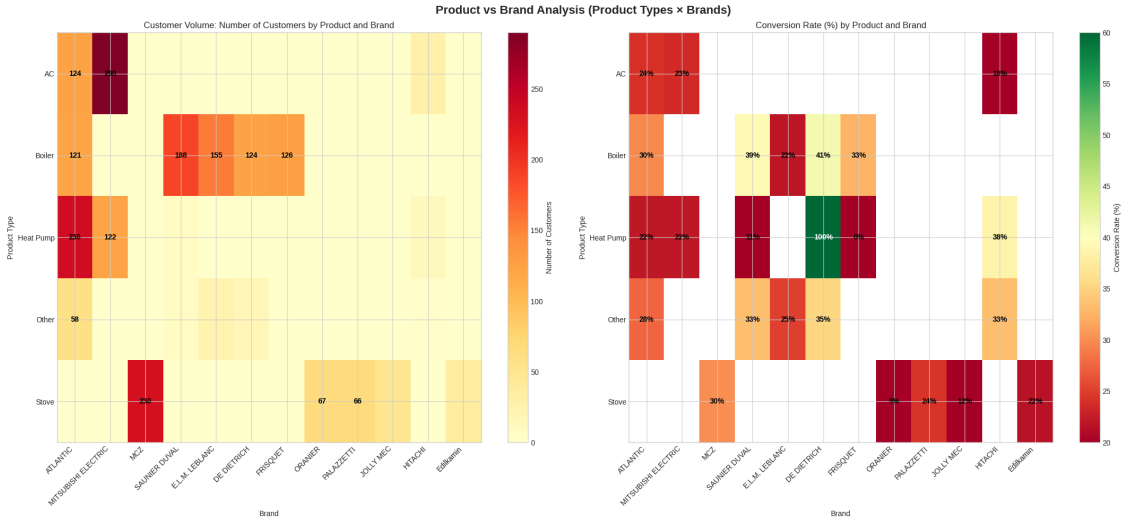



```
Analyzing 2,310 customers
Price range: €1,327 to €25,676
Median price: €6,126

================================================================================
DYNAMIC INSIGHTS CALCULATED:
================================================================================
Sweet spots: €4.4k (44.2%) and €3.6k (40.3%)
Dead zone: €6.7k-7.0k (14-21%)
Heat pumps: 77% subsidy issues, 23.7% conversion
Price sensitivity: Stove (22.5pts) > Heat Pump (14.9pts) > Boiler (14.7pts) > AC
(7.8pts)
Same day decisions: 88%
```




```
Actual product categories in data: ['AC', 'Boiler', 'Heat Pump', 'Other',
'Stove']
```



```
================================================================================
PRODUCT × BRAND ANALYSIS
================================================================================

  CUSTOMER VOLUME (Number of Customers):
--------------------------------------------------------------------------------
main_brand                ATLANTIC  MITSUBISHI ELECTRIC  MCZ  SAUNIER DUVAL
E.L.M. LEBLANC  DE DIETRICH  FRISQUET  ORANIER  PALAZZETTI  JOLLY MEC  HITACHI
Edilkamin
main_equipment_category
AC                            124                  290    0              0
0            0         0        0           0          0       28          0
Boiler                        121                    0    0            188
155          124       126        0           0          0        0          0
Heat Pump                     236                  122    0              9
0            1         1        0           0          0       13          0
Other                          58                    0    0              9
24           17         0        0           0          0        3          0
Stove                           0                    0  230              0
0            0         0       67          66         49        0         37

================================================================================
  CONVERSION RATES (%):
--------------------------------------------------------------------------------
main_brand                ATLANTIC  MITSUBISHI ELECTRIC   MCZ  SAUNIER DUVAL
E.L.M. LEBLANC  DE DIETRICH  FRISQUET  ORANIER  PALAZZETTI  JOLLY MEC  HITACHI
Edilkamin
main_equipment_category
AC                           24.2                 23.4   NaN            NaN
NaN          NaN       NaN      NaN         NaN        NaN     17.9        NaN
Boiler                       29.8                  NaN   NaN           39.4
21.9         41.1      32.5      NaN         NaN        NaN      NaN        NaN
Heat Pump                    22.0                 22.1   NaN           11.1
NaN        100.0       0.0      NaN         NaN        NaN     38.5        NaN
Other                        27.6                  NaN   NaN           33.3
25.0         35.3       NaN      NaN         NaN        NaN     33.3        NaN
Stove                         NaN                  NaN  30.0            NaN
NaN          NaN       NaN      9.0        24.2       12.2      NaN       21.6

================================================================================
  BEST BRAND BY PRODUCT TYPE
--------------------------------------------------------------------------------

AC:
```

```
  Best brand: ATLANTIC
  Conversion: 24.2%
  Customers: 124
  Top 3 brands:
    1. ATLANTIC: 24.2% (n=124)
    2. MITSUBISHI ELECTRIC: 23.4% (n=290)
    3. HITACHI: 17.9% (n=28)

Boiler:
  Best brand: DE DIETRICH
  Conversion: 41.1%
  Customers: 124
  Top 3 brands:
    1. DE DIETRICH: 41.1% (n=124)
    2. SAUNIER DUVAL: 39.4% (n=188)
    3. FRISQUET: 32.5% (n=126)

Heat Pump:
  Best brand: DE DIETRICH
  Conversion: 100.0%
  Customers: 1
  Top 3 brands:
    1. DE DIETRICH: 100.0% (n=1)
    2. HITACHI: 38.5% (n=13)
    3. MITSUBISHI ELECTRIC: 22.1% (n=122)

Other:
  Best brand: DE DIETRICH
  Conversion: 35.3%
  Customers: 17
  Top 3 brands:
    1. DE DIETRICH: 35.3% (n=17)
    2. HITACHI: 33.3% (n=3)
    3. SAUNIER DUVAL: 33.3% (n=9)

Stove:
  Best brand: MCZ
  Conversion: 30.0%
  Customers: 230
  Top 3 brands:
    1. MCZ: 30.0% (n=230)
    2. PALAZZETTI: 24.2% (n=66)
    3. Edilkamin: 21.6% (n=37)

================================================================================
 BEST PRODUCT BY BRAND
--------------------------------------------------------------------------------
```

```
ATLANTIC:
  Best product: Boiler
  Conversion: 29.8%
  Customers: 121
  All products:
    Boiler: 29.8% (n=121)
    Other: 27.6% (n=58)
    AC: 24.2% (n=124)
    Heat Pump: 22.0% (n=236)

MITSUBISHI ELECTRIC:
  Best product: AC
  Conversion: 23.4%
  Customers: 290
  All products:
    AC: 23.4% (n=290)
    Heat Pump: 22.1% (n=122)

MCZ:
  Best product: Stove
  Conversion: 30.0%
  Customers: 230
  All products:
    Stove: 30.0% (n=230)

SAUNIER DUVAL:
  Best product: Boiler
  Conversion: 39.4%
  Customers: 188
  All products:
    Boiler: 39.4% (n=188)
    Other: 33.3% (n=9)
    Heat Pump: 11.1% (n=9)

E.L.M. LEBLANC:
  Best product: Other
  Conversion: 25.0%
  Customers: 24
  All products:
    Other: 25.0% (n=24)
    Boiler: 21.9% (n=155)

DE DIETRICH:
  Best product: Heat Pump
  Conversion: 100.0%
  Customers: 1
  All products:
    Heat Pump: 100.0% (n=1)
```

```
    Boiler: 41.1% (n=124)
    Other: 35.3% (n=17)

FRISQUET:
  Best product: Boiler
  Conversion: 32.5%
  Customers: 126
  All products:
    Boiler: 32.5% (n=126)
    Heat Pump: 0.0% (n=1)

ORANIER:
  Best product: Stove
  Conversion: 9.0%
  Customers: 67
  All products:
    Stove: 9.0% (n=67)

================================================================================
 STRATEGIC INSIGHTS
================================================================================

Brand Dominance by Product Category:

AC:
  Volume leader: MITSUBISHI ELECTRIC (290 customers, 65.6% share)
  Conversion leader: ATLANTIC (24.2%)

Boiler:
  Volume leader: SAUNIER DUVAL (188 customers, 26.3% share)
  Conversion leader: DE DIETRICH (41.1%)

Heat Pump:
  Volume leader: ATLANTIC (236 customers, 61.8% share)
  Conversion leader: DE DIETRICH (100.0%)

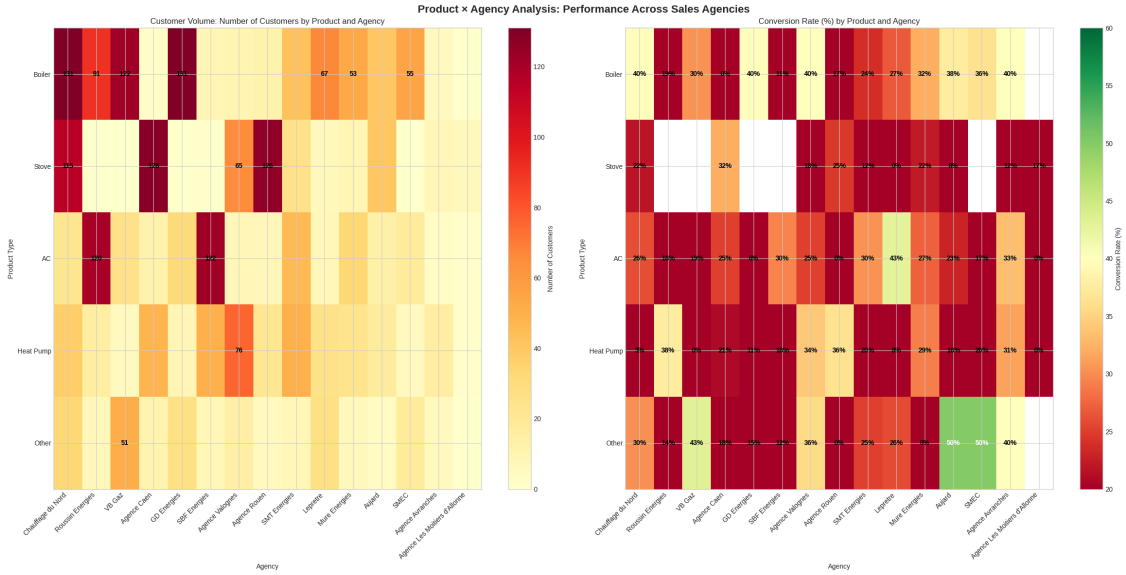
Other:
  Volume leader: ATLANTIC (58 customers, 52.3% share)
  Conversion leader: DE DIETRICH (35.3%)

Stove:
  Volume leader: MCZ (230 customers, 51.2% share)
  Conversion leader: MCZ (30.0%)

Actual product categories in data: ['Boiler', 'Stove', 'AC', 'Heat Pump',
'Other']

Top 15 agencies by customer volume:
```

```
Chauffage du Nord: 339 customers, 27.7% conversion
Roussin Energies: 234 customers, 19.7% conversion
VB Gaz: 204 customers, 31.4% conversion
Agence Caen: 201 customers, 27.9% conversion
GD Energies: 198 customers, 29.8% conversion
SBF Energies: 189 customers, 24.9% conversion
Agence Valognes: 173 customers, 28.3% conversion
Agence Rouen: 172 customers, 23.8% conversion
SMT Energies: 172 customers, 22.7% conversion
Lepretre: 132 customers, 22.7% conversion
Mure Energies: 125 customers, 28.0% conversion
Aujard: 114 customers, 22.8% conversion
SMEC: 96 customers, 34.4% conversion
Agence Avranches: 37 customers, 29.7% conversion
Agence Les Moitiers d'Allonne: 8 customers, 12.5% conversion
```




```
================================================================================
PRODUCT × AGENCY ANALYSIS
================================================================================

  CUSTOMER VOLUME BY PRODUCT AND AGENCY:
--------------------------------------------------------------------------------
--------------------
main_agency                Chauffage du Nord  Roussin Energies  VB Gaz  Agence
Caen  GD Energies  SBF Energies  Agence Valognes  Agence Rouen  SMT Energies
Lepretre  Mure Energies  Aujard  SMEC  Agence Avranches  Agence Les Moitiers
d'Allonne
main_equipment_category
```

```
Boiler                                  131                91     122
2          131             9               10            12            42
67             53      40    55                 5                                0
Stove                                   115                 0       0
128            0             0               65           126            26
6              9      40     0                 8                                6
AC                                       23               120      26
12           31           122                8             7            46
7             33      13    18                 3                                1
Heat Pump                                37                16       5
48            9            50               76            22            50
25             24      17     5                16                                1
Other                                    33                 7      51
11           27             8               14             5             8
27              6       4    18                 5                                0

==============================================================================
 CONVERSION RATES (%) BY PRODUCT AND AGENCY:
------------------------------------------------------------------------------
--------------------
main_agency              Chauffage du Nord  Roussin Energies  VB Gaz  Agence
Caen  GD Energies  SBF Energies  Agence Valognes  Agence Rouen  SMT Energies
Lepretre  Mure Energies  Aujard  SMEC  Agence Avranches  Agence Les Moitiers
d'Allonne
main_equipment_category
Boiler                                  39.7              18.7    30.3
0.0         39.7          11.1             40.0          16.7          23.8
26.9           32.1    37.5  36.4              40.0
NaN
Stove                                   21.7               NaN     NaN
32.0          NaN           NaN             18.5          24.6          11.5
0.0           22.2     7.5   NaN              12.5
16.7
AC                                      26.1              18.3    19.2
25.0          6.5          29.5             25.0           0.0          30.4
42.9           27.3    23.1  16.7              33.3
0.0
Heat Pump                                2.7              37.5     0.0
20.8         11.1          18.0             34.2          36.4          20.0
8.0           29.2    17.6  20.0              31.2
0.0
Other                                   30.3              14.3    43.1
18.2         14.8          12.5             35.7           0.0          25.0
25.9            0.0    50.0  50.0              40.0
NaN

==============================================================================
 BEST AGENCY BY PRODUCT TYPE
```

```
--------------------------------------------------------------------------------
--------------------

Boiler:
  Best agency: Agence Avranches
  Conversion: 40.0%
  Customers: 5
  Top 3 agencies:
    1. Agence Avranches: 40.0% (n=5)
    2. Agence Valognes: 40.0% (n=10)
    3. Chauffage du Nord: 39.7% (n=131)

Stove:
  Best agency: Agence Caen
  Conversion: 32.0%
  Customers: 128
  Top 3 agencies:
    1. Agence Caen: 32.0% (n=128)
    2. Agence Rouen: 24.6% (n=126)
    3. Mure Energies: 22.2% (n=9)

AC:
  Best agency: Lepretre
  Conversion: 42.9%
  Customers: 7
  Top 3 agencies:
    1. Lepretre: 42.9% (n=7)
    2. Agence Avranches: 33.3% (n=3)
    3. SMT Energies: 30.4% (n=46)

Heat Pump:
  Best agency: Roussin Energies
  Conversion: 37.5%
  Customers: 16
  Top 3 agencies:
    1. Roussin Energies: 37.5% (n=16)
    2. Agence Rouen: 36.4% (n=22)
    3. Agence Valognes: 34.2% (n=76)

Other:
  Best agency: Aujard
  Conversion: 50.0%
  Customers: 4
  Top 3 agencies:
    1. Aujard: 50.0% (n=4)
    2. SMEC: 50.0% (n=18)
    3. VB Gaz: 43.1% (n=51)
```

```
================================================================================
  BEST PRODUCT BY AGENCY
--------------------------------------------------------------------------------
--------------------

Chauffage du Nord:
  Best product: Boiler
  Conversion: 39.7%
  Customers: 131
  All products:
    Boiler: 39.7% (n=131)
    Other: 30.3% (n=33)
    AC: 26.1% (n=23)
    Stove: 21.7% (n=115)
    Heat Pump: 2.7% (n=37)

Roussin Energies:
  Best product: Heat Pump
  Conversion: 37.5%
  Customers: 16
  All products:
    Heat Pump: 37.5% (n=16)
    Boiler: 18.7% (n=91)
    AC: 18.3% (n=120)
    Other: 14.3% (n=7)

VB Gaz:
  Best product: Other
  Conversion: 43.1%
  Customers: 51
  All products:
    Other: 43.1% (n=51)
    Boiler: 30.3% (n=122)
    AC: 19.2% (n=26)
    Heat Pump: 0.0% (n=5)

Agence Caen:
  Best product: Stove
  Conversion: 32.0%
  Customers: 128
  All products:
    Stove: 32.0% (n=128)
    AC: 25.0% (n=12)
    Heat Pump: 20.8% (n=48)
    Other: 18.2% (n=11)
    Boiler: 0.0% (n=2)

GD Energies:
```

```
  Best product: Boiler
  Conversion: 39.7%
  Customers: 131
  All products:
    Boiler: 39.7% (n=131)
    Other: 14.8% (n=27)
    Heat Pump: 11.1% (n=9)
    AC: 6.5% (n=31)

SBF Energies:
  Best product: AC
  Conversion: 29.5%
  Customers: 122
  All products:
    AC: 29.5% (n=122)
    Heat Pump: 18.0% (n=50)
    Other: 12.5% (n=8)
    Boiler: 11.1% (n=9)

Agence Valognes:
  Best product: Boiler
  Conversion: 40.0%
  Customers: 10
  All products:
    Boiler: 40.0% (n=10)
    Other: 35.7% (n=14)
    Heat Pump: 34.2% (n=76)
    AC: 25.0% (n=8)
    Stove: 18.5% (n=65)

Agence Rouen:
  Best product: Heat Pump
  Conversion: 36.4%
  Customers: 22
  All products:
    Heat Pump: 36.4% (n=22)
    Stove: 24.6% (n=126)
    Boiler: 16.7% (n=12)
    AC: 0.0% (n=7)
    Other: 0.0% (n=5)

SMT Energies:
  Best product: AC
  Conversion: 30.4%
  Customers: 46
  All products:
    AC: 30.4% (n=46)
    Other: 25.0% (n=8)
```

```
    Boiler: 23.8% (n=42)
    Heat Pump: 20.0% (n=50)
    Stove: 11.5% (n=26)

Lepretre:
  Best product: AC
  Conversion: 42.9%
  Customers: 7
  All products:
    AC: 42.9% (n=7)
    Boiler: 26.9% (n=67)
    Other: 25.9% (n=27)
    Heat Pump: 8.0% (n=25)
    Stove: 0.0% (n=6)

================================================================================
 AGENCY PERFORMANCE SUMMARY
================================================================================

Top 15 agencies by volume:
                              conversion_rate  customer_count  heat_pump_pct
subsidy_issue_pct  avg_price
main_agency
Chauffage du Nord                        27.7             339           11.5
40.7   6447.915
Roussin Energies                         19.7             234           10.7
9.8   7577.820
VB Gaz                                   31.4             204            2.5
1.5   5072.292
Agence Caen                              27.9             201           25.9
33.3   8808.933
GD Energies                              29.8             198            7.6
5.1   5287.781
SBF Energies                             24.9             189           32.8
25.4   9305.760
Agence Valognes                          28.3             173           46.2
43.4   9905.586
SMT Energies                             22.7             172           33.7
50.6   8597.431
Agence Rouen                             23.8             172           12.8
36.6   7527.070
Lepretre                                 22.7             132           19.7
13.6   7344.482
Mure Energies                            28.0             125           22.4
21.6   6432.522
Aujard                                   22.8             114           15.8
14.0   8440.772
SMEC                                     34.4              96            5.2
```

```
12.5   6203.087
Agence Avranches                          29.7              37           45.9
56.8  10866.670
Agence Les Moitiers d'Allonne             12.5               8           12.5
12.5   8253.901

================================================================================
  STRATEGIC INSIGHTS
================================================================================

Agency Excellence by Product Category:

Boiler:
  Conversion leader: Agence Valognes (40.0%, n=10)

Stove:
  Conversion leader: Agence Caen (32.0%, n=128)

AC:
  Conversion leader: Lepretre (42.9%, n=7)

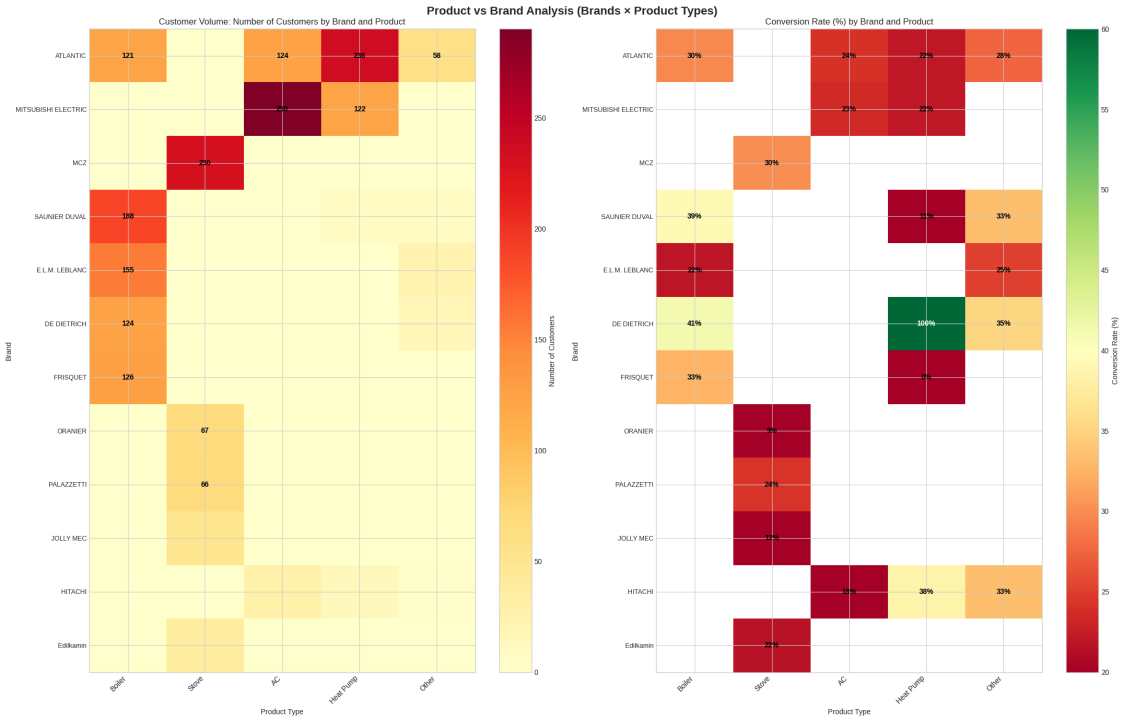
Heat Pump:
  Conversion leader: Roussin Energies (37.5%, n=16)

Other:
  Conversion leader: Aujard (50.0%, n=4)

================================================================================
  HEAT PUMP SPECIALISTS
================================================================================

Top 5 agencies for Heat Pump conversion:
  Roussin Energies: 37.5% (n=16)
  Agence Rouen: 36.4% (n=22)
  Agence Valognes: 34.2% (n=76)
  Agence Avranches: 31.2% (n=16)
  Mure Energies: 29.2% (n=24)

Actual product categories in data: ['Boiler', 'Stove', 'AC', 'Heat Pump',
'Other']
```




```
================================================================================
BRAND × PRODUCT ANALYSIS (Swapped Axes)
================================================================================

  CUSTOMER VOLUME (Number of Customers):
--------------------------------------------------------------------------------
main_equipment_category  Boiler  Stove   AC  Heat Pump  Other
main_brand
ATLANTIC                   121      0  124        236     58
MITSUBISHI ELECTRIC          0      0  290        122      0
MCZ                          0    230    0          0      0
SAUNIER DUVAL              188      0    0          9      9
E.L.M. LEBLANC             155      0    0          0     24
DE DIETRICH                124      0    0          1     17
FRISQUET                   126      0    0          1      0
ORANIER                      0     67    0          0      0
PALAZZETTI                   0     66    0          0      0
JOLLY MEC                    0     49    0          0      0
HITACHI                      0      0   28         13      3
Edilkamin                    0     37    0          0      0

================================================================================
  CONVERSION RATES (%):
```

```
--------------------------------------------------------------------------------
main_equipment_category  Boiler  Stove    AC  Heat Pump  Other
main_brand
ATLANTIC                   29.8    NaN  24.2       22.0   27.6
MITSUBISHI ELECTRIC         NaN    NaN  23.4       22.1    NaN
MCZ                         NaN   30.0   NaN        NaN    NaN
SAUNIER DUVAL              39.4    NaN   NaN       11.1   33.3
E.L.M. LEBLANC             21.9    NaN   NaN        NaN   25.0
DE DIETRICH                41.1    NaN   NaN      100.0   35.3
FRISQUET                   32.5    NaN   NaN        0.0    NaN
ORANIER                     NaN    9.0   NaN        NaN    NaN
PALAZZETTI                  NaN   24.2   NaN        NaN    NaN
JOLLY MEC                   NaN   12.2   NaN        NaN    NaN
HITACHI                     NaN    NaN  17.9       38.5   33.3
Edilkamin                   NaN   21.6   NaN        NaN    NaN

================================================================================
 BEST PRODUCT BY BRAND
--------------------------------------------------------------------------------

ATLANTIC:
  Best product: Boiler
  Conversion: 29.8%
  Customers: 121
  Top 3 products:
    1. Boiler: 29.8% (n=121)
    2. Other: 27.6% (n=58)
    3. AC: 24.2% (n=124)

MITSUBISHI ELECTRIC:
  Best product: AC
  Conversion: 23.4%
  Customers: 290
  Top 3 products:
    1. AC: 23.4% (n=290)
    2. Heat Pump: 22.1% (n=122)

MCZ:
  Best product: Stove
  Conversion: 30.0%
  Customers: 230
  Top 3 products:
    1. Stove: 30.0% (n=230)

SAUNIER DUVAL:
  Best product: Boiler
  Conversion: 39.4%
  Customers: 188
```

```
  Top 3 products:
    1. Boiler: 39.4% (n=188)
    2. Other: 33.3% (n=9)
    3. Heat Pump: 11.1% (n=9)

E.L.M. LEBLANC:
  Best product: Other
  Conversion: 25.0%
  Customers: 24
  Top 3 products:
    1. Other: 25.0% (n=24)
    2. Boiler: 21.9% (n=155)

DE DIETRICH:
  Best product: Heat Pump
  Conversion: 100.0%
  Customers: 1
  Top 3 products:
    1. Heat Pump: 100.0% (n=1)
    2. Boiler: 41.1% (n=124)
    3. Other: 35.3% (n=17)

FRISQUET:
  Best product: Boiler
  Conversion: 32.5%
  Customers: 126
  Top 3 products:
    1. Boiler: 32.5% (n=126)
    2. Heat Pump: 0.0% (n=1)

ORANIER:
  Best product: Stove
  Conversion: 9.0%
  Customers: 67
  Top 3 products:
    1. Stove: 9.0% (n=67)

PALAZZETTI:
  Best product: Stove
  Conversion: 24.2%
  Customers: 66
  Top 3 products:
    1. Stove: 24.2% (n=66)

JOLLY MEC:
  Best product: Stove
  Conversion: 12.2%
  Customers: 49
```

```
  Top 3 products:
    1. Stove: 12.2% (n=49)

================================================================================
 BEST BRAND BY PRODUCT TYPE
--------------------------------------------------------------------------------

Boiler:
  Best brand: DE DIETRICH
  Conversion: 41.1%
  Customers: 124
  Top 5 brands:
    1. DE DIETRICH: 41.1% (n=124)
    2. SAUNIER DUVAL: 39.4% (n=188)
    3. FRISQUET: 32.5% (n=126)
    4. ATLANTIC: 29.8% (n=121)
    5. E.L.M. LEBLANC: 21.9% (n=155)

Stove:
  Best brand: MCZ
  Conversion: 30.0%
  Customers: 230
  Top 5 brands:
    1. MCZ: 30.0% (n=230)
    2. PALAZZETTI: 24.2% (n=66)
    3. Edilkamin: 21.6% (n=37)
    4. JOLLY MEC: 12.2% (n=49)
    5. ORANIER: 9.0% (n=67)

AC:
  Best brand: ATLANTIC
  Conversion: 24.2%
  Customers: 124
  Top 5 brands:
    1. ATLANTIC: 24.2% (n=124)
    2. MITSUBISHI ELECTRIC: 23.4% (n=290)
    3. HITACHI: 17.9% (n=28)

Heat Pump:
  Best brand: DE DIETRICH
  Conversion: 100.0%
  Customers: 1
  Top 5 brands:
    1. DE DIETRICH: 100.0% (n=1)
    2. HITACHI: 38.5% (n=13)
    3. MITSUBISHI ELECTRIC: 22.1% (n=122)
    4. ATLANTIC: 22.0% (n=236)
    5. SAUNIER DUVAL: 11.1% (n=9)
```

```
Other:
  Best brand: DE DIETRICH
  Conversion: 35.3%
  Customers: 17
  Top 5 brands:
    1. DE DIETRICH: 35.3% (n=17)
    2. HITACHI: 33.3% (n=3)
    3. SAUNIER DUVAL: 33.3% (n=9)
    4. ATLANTIC: 27.6% (n=58)
    5. E.L.M. LEBLANC: 25.0% (n=24)

================================================================================
 STRATEGIC INSIGHTS
================================================================================

Brand Performance by Product Category:

ATLANTIC:
  Boiler: 29.8% (n=121)
  Other: 27.6% (n=58)
  AC: 24.2% (n=124)

MITSUBISHI ELECTRIC:
  AC: 23.4% (n=290)
  Heat Pump: 22.1% (n=122)

MCZ:
  Stove: 30.0% (n=230)

SAUNIER DUVAL:
  Boiler: 39.4% (n=188)
  Other: 33.3% (n=9)
  Heat Pump: 11.1% (n=9)

E.L.M. LEBLANC:
  Other: 25.0% (n=24)
  Boiler: 21.9% (n=155)

DE DIETRICH:
  Heat Pump: 100.0% (n=1)
  Boiler: 41.1% (n=124)
  Other: 35.3% (n=17)

FRISQUET:
  Boiler: 32.5% (n=126)
  Heat Pump: 0.0% (n=1)
```

```
ORANIER:
  Stove: 9.0% (n=67)

================================================================================
 MARKET LEADERS BY PRODUCT CATEGORY
================================================================================

Boiler:
  Volume leader: SAUNIER DUVAL (188 customers, 26.3% share)
  Conversion leader: DE DIETRICH (41.1%)

Stove:
  Volume leader: MCZ (230 customers, 51.2% share)
  Conversion leader: MCZ (30.0%)

AC:
  Volume leader: MITSUBISHI ELECTRIC (290 customers, 65.6% share)
  Conversion leader: ATLANTIC (24.2%)

Heat Pump:
  Volume leader: ATLANTIC (236 customers, 61.8% share)
  Conversion leader: DE DIETRICH (100.0%)

Other:
  Volume leader: ATLANTIC (58 customers, 52.3% share)
  Conversion leader: DE DIETRICH (35.3%)

 (main_equipment_category  Boiler  Stove   AC  Heat Pump  Other
  main_brand
  ATLANTIC                    121      0  124        236     58
  MITSUBISHI ELECTRIC           0      0  290        122      0
  MCZ                           0    230    0          0      0
  SAUNIER DUVAL               188      0    0          9      9
  E.L.M. LEBLANC              155      0    0          0     24
  DE DIETRICH                 124      0    0          1     17
  FRISQUET                    126      0    0          1      0
  ORANIER                       0     67    0          0      0
  PALAZZETTI                    0     66    0          0      0
  JOLLY MEC                     0     49    0          0      0
  HITACHI                       0      0   28         13      3
  Edilkamin                     0     37    0          0      0,
  main_equipment_category     Boiler      Stove         AC   Heat Pump  \
  main_brand
  ATLANTIC                 29.752066        NaN  24.193548   22.033898
  MITSUBISHI ELECTRIC            NaN        NaN  23.448276   22.131148
  MCZ                            NaN  30.000000        NaN         NaN
  SAUNIER DUVAL            39.361702        NaN        NaN   11.111111
```

```
E.L.M. LEBLANC           21.935484        NaN        NaN         NaN
DE DIETRICH              41.129032        NaN        NaN  100.000000
FRISQUET                 32.539683        NaN        NaN    0.000000
ORANIER                        NaN   8.955224        NaN         NaN
PALAZZETTI                     NaN  24.242424        NaN         NaN
JOLLY MEC                      NaN  12.244898        NaN         NaN
HITACHI                        NaN        NaN  17.857143   38.461538
Edilkamin                      NaN  21.621622        NaN         NaN

main_equipment_category      Other
main_brand
ATLANTIC                 27.586207
MITSUBISHI ELECTRIC            NaN
MCZ                            NaN
SAUNIER DUVAL            33.333333
E.L.M. LEBLANC           25.000000
DE DIETRICH              35.294118
FRISQUET                       NaN
ORANIER                        NaN
PALAZZETTI                     NaN
JOLLY MEC                      NaN
HITACHI                  33.333333
Edilkamin                      NaN  )
```

```python
import ipynbname
import subprocess
import os
import sys

def export_notebook_to_pdf(overwrite=True, verbose=True):
    """
    Export current notebook to PDF with overwrite control

    Args:
        overwrite (bool): If True, overwrite existing PDF file
        verbose (bool): If True, print progress messages

    Returns:
        bool: True if successful, False otherwise
    """
    try:
        # Get notebook path
        notebook_path = ipynbname.path()
        notebook_path_str = str(notebook_path)
        pdf_path = notebook_path_str.replace('.ipynb', '.pdf')

        if verbose:
```

```
            print(f"  Notebook: {notebook_path.name}")
            print(f"  PDF path: {pdf_path}")

        # Check if PDF exists
        if os.path.exists(pdf_path):
            if overwrite:
                if verbose:
                    print(f"  Removing existing PDF: {pdf_path}")
                os.remove(pdf_path)
            else:
                if verbose:
                    print(f"  PDF already exists and overwrite=False")
                return False

        # Convert to PDF
        if verbose:
            print("  Converting to PDF...")

        result = subprocess.run([
            'jupyter', 'nbconvert', '--to', 'pdf',
            '--no-prompt',  # Suppress input prompts
            notebook_path_str
        ], capture_output=True, text=True)

        if result.returncode == 0:
            if verbose:
                print(f"  PDF saved: {pdf_path}")
            return True
        else:
            if verbose:
                print(f"  Conversion failed:")
                print(result.stderr)
            return False

    except Exception as e:
        if verbose:
            print(f"  Error: {e}")
        return False

# Use the function
if __name__ == "__main__":
    # Overwrite existing PDF
    export_notebook_to_pdf(overwrite=True)

    # Or create new version without overwriting
    # export_notebook_to_pdf(overwrite=False)export_full_notebook_to_pdf()
```

```
Error: Can't identify the notebook path.
```

```python
import ipynbname

try:
    print(f"Path: {ipynbname.path()}")
    print(f"Name: {ipynbname.name()}")
except Exception as e:
    print(f"Error: {e}")
    print(f"Error type: {type(e)}")
```

```
Error: Can't identify the notebook path.
Error type: <class 'FileNotFoundError'>
```

```python
# Restart kernel, then run this first cell alone
import ipynbname
print(ipynbname.path())  # Should work after fresh run
```

```
---------------------------------------------------------------------------
FileNotFoundError                         Traceback (most recent call last)
Cell In[11], line 3
      1 # Restart kernel, then run this first cell alone
      2 import ipynbname
----> 3 print(ipynbname.path())  # Should work after fresh run

File ~/project/homeserve/credit_policy/.direnv/python-3.12.0/lib/python3.12/
 ↪site-packages/ipynbname/__init__.py:120, in path()
    117 if path:
    118     return path
--> 120 raise FileNotFoundError(FILE_ERROR.format('path'))

FileNotFoundError: Can't identify the notebook path.
```

```python
from IPython.display import display, Javascript
import json

# This gets the notebook name from frontend
javascript = """
var kernel = IPython.notebook.kernel;
var notebook_name = IPython.notebook.notebook_name;
kernel.execute("notebook_name = '" + notebook_name + "'");
"""
display(Javascript(javascript))
print(f"Notebook name: {notebook_name}")  # Should work now
```

```
<IPython.core.display.Javascript object>
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
```

```
Cell In[12], line 11
      5 javascript = """
      6 var kernel = IPython.notebook.kernel;
      7 var notebook_name = IPython.notebook.notebook_name;
      8 kernel.execute("notebook_name = '" + notebook_name + "'");
      9 """
     10 display(Javascript(javascript))
---> 11 print(f"Notebook name: {notebook_name}")  # Should work now

NameError: name 'notebook_name' is not defined
```